

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

Course Code: ITA05	Course Title: COMPUTER VISION
CO Assessed: CO2 – Manipulation (BL3)	Assessment: Assessment Tool 1 – Analytical Problem Solving
Weightage: 20% of CIA	Total Marks: 50 (5 Questions × 10 Marks)

CO2 Statement: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BL3)

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Section / Batch: Slot C / 2023	Date: 21/08/26

Code of Conduct

I (DK Hari Krishna / 192321176) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: DK Hari Krishna

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic Focus	Q Nos	Marks
Grey Level Transformations Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10

GRAND TOTAL: 50 Marks

Annexure A — Analytical Problem Solving: Answers

Note: Where the original question's numeric pixel values / kernel were not legible in the source file, standard representative values consistent with the stated problem type have been used below to demonstrate the complete method — substitute your own given values into the same formulas and steps.

Q1. Grey Level Transformation — Negative Transformation

(a) Purpose of Grey Level Transformation

Grey level (intensity) transformations map each input pixel intensity r to an output intensity s using a function $s = T(r)$, applied independently to every pixel. Their purpose is to enhance visual quality — improving contrast, revealing hidden detail in very dark or very bright regions, correcting poor illumination, or preparing an image for further processing (segmentation, feature extraction) — without altering the spatial relationships between pixels.

(b) Negative Transformation

The negative transformation is defined as:

$$s = (L - 1) - r$$

where r is the input intensity, s is the output intensity, and L is the number of grey levels ($L = 256$ for an 8-bit image, so $L - 1 = 255$). This reverses the intensity scale: dark pixels become light and light pixels become dark, producing a photographic-negative effect.

Applying $s = 255 - r$ to a representative set of pixel values:

r (input)	0	50	100	150	200	255
$s = 255 - r$	255	205	155	105	55	0

Interpretation

- Every pixel value is subtracted from 255, so the intensity scale is flipped end-to-end ($0 \leftrightarrow 255$, $50 \leftrightarrow 205$, etc.).
- Regions that were dark (low r) become bright (high s), and regions that were bright become dark.
- The transformation is particularly useful for enhancing white or grey detail embedded in large dark regions of an image — for example, revealing fine structures in dark portions of medical X-ray or mammogram images — since the negative makes such detail appear bright and easier to inspect.
- Because the mapping is linear and one-to-one, no information is lost; the transformation is fully reversible by applying the same operation again.

Q2. Histogram Equalization

(a) Why Histogram Equalization Is Used

Histogram equalization is a contrast-enhancement technique that redistributes the intensity values of an image so that the output histogram is approximately uniform. Images with poor contrast typically have pixel intensities concentrated in a narrow range; by spreading these values across the full available range, histogram equalization increases the dynamic range and makes previously indistinguishable details visible, improving overall visual contrast automatically without manual parameter tuning.

(b) Computing the Cumulative Distribution Function (CDF)

This is a 4-bit grayscale image, so $L = 16$ possible intensity levels (0–15). Given intensities 0–3 are the only levels present, with total pixel count $MN = 4 + 6 + 10 + 6 = 26$. The PDF (normalized histogram) and CDF are:

Intensity (r_k)	0	1	2	3
Frequency (n_k)	4	6	10	6
PDF $p(r_k) = n_k / MN$	0.1538	0.2308	0.3846	0.2308
CDF	0.1538	0.3846	0.7692	1.0000

(c) Equalized Intensity Values

The equalized output level is computed as:

$$s_k = \text{round}[(L - 1) \times \text{CDF}(r_k)] \quad , \quad L - 1 = 15$$

r_k	0	1	2	3
CDF	0.1538	0.3846	0.7692	1.0000
$s_k = \text{round}(15 \times \text{CDF})$	2	6	12	15

Interpretation

- The original intensities (0, 1, 2, 3) — all crowded into a narrow 4-level band — are mapped to the equalized values (2, 6, 12, 15), which are spread across nearly the full 0–15 range.
- This expansion means small differences between intensity levels in the original image now correspond to much larger, more visible differences in the equalized output, directly improving perceived contrast.
- Level 2, which had the highest frequency (10 pixels, the dominant mid-tone), maps close to the middle-upper end of the new range, reflecting its cumulative weight in the distribution.

Q3. Gaussian Smoothing

(a) Why Gaussian Filtering Is Preferred Over Averaging

A simple averaging (box) filter assigns equal weight to every pixel in the neighborhood, which blurs edges uniformly and can produce visible ringing/blocky artifacts, since it treats near and far neighbors identically. A Gaussian filter instead weights pixels according to their distance from the center using a bell-shaped (Gaussian) distribution — closer pixels contribute more, farther pixels contribute less. This produces smoother, more natural-looking blurring, better preserves edge structure, avoids the sharp cut-off artifacts of a box filter, and — being separable and rotationally symmetric — introduces no directional bias in the smoothing.

(b) Applying the Gaussian Kernel

Using the standard normalized 3×3 Gaussian kernel ($\sigma \approx 1$):

1	2	1
2	4	2
1	2	1

(kernel normalized by $1/16$, i.e. divide the weighted sum by 16)

to the image region:

100	102	104
108	150	110
112	114	116

The new center pixel value is the weighted sum of the region divided by the sum of kernel weights (16):

$$\text{New center} = [\sum (\text{pixel} \times \text{weight})] / 16 = 1900 / 16 = 118.75 \approx 119$$

Interpretation

- The Gaussian-smoothed center value (119) is noticeably lower than the original noisy center pixel (150), showing that the filter has suppressed the local intensity spike — i.e., reduced noise.
- For comparison, a simple 3×3 averaging filter (equal weights of $1/9$) on the same region gives $1016 / 9 = 112.89 \approx 113$. Both filters smooth the region, but the Gaussian result stays slightly closer to the true local structure because it weights the center and near neighbors more heavily than the equal-weighted average.
- This confirms the general principle: Gaussian smoothing reduces random noise while better preserving meaningful local detail than uniform averaging.

Q4. Edge Detection Using the Sobel Operator

(a) Concept and Purpose of Edge Detection

Edge detection identifies locations in an image where intensity changes sharply — these correspond to object boundaries, shape outlines, and significant structural features. It works by computing the image gradient (rate of intensity change) in the horizontal and/or vertical direction; a large gradient magnitude indicates a strong edge. Edge detection is a key preprocessing step for segmentation, object recognition, and feature extraction, since it reduces the image to its most structurally significant information while discarding redundant, near-uniform regions.

(b) Applying the Horizontal Sobel Mask

The horizontal Sobel mask G_x (which responds strongly to vertical edges, i.e. left–right intensity transitions) is:

-1	0	1
-2	0	2
-1	0	1

Applied to the image region:

10	10	100
10	10	100
10	10	100

The horizontal gradient at the center pixel is the sum of the element-wise products of the mask and the region:

$$G_x = \sum (pixel \times mask\ weight) = 360$$

Interpretation

- The large gradient magnitude ($G_x = 360$) indicates a strong intensity transition at this location — the region contains a clear vertical edge, with dark pixels (value 10) on the left and bright pixels (value 100) on the right.
- A gradient close to zero would indicate a flat, uniform region with no edge, whereas this large positive value confirms an edge running vertically through the center of the neighborhood.
- In practice, this G_x value would be combined with the vertical Sobel response G_y (using $|G| = \sqrt{G_x^2 + G_y^2}$, or the simpler approximation $|G_x| + |G_y|$) to obtain the overall edge strength and, if needed, thresholded to produce a binary edge map.

Q5. Contrast Stretching

(a) Need for Contrast Stretching

An image whose intensity values occupy only a narrow band (here, 50–150 out of the full possible 0–255 range) appears washed out or low-contrast, since the visual difference between adjacent grey levels is small. Contrast stretching (a linear grey-level transformation) expands this narrow range to occupy the full dynamic range of the display, making subtle intensity differences far more perceptible and significantly improving visual quality without changing the relative ordering of pixel values.

(b) Linear Contrast Stretching

Mapping the original range [50, 150] to the full range [0, 255] using the linear formula:

$$s = [(r - rmin) / (rmax - rmin)] \times (smax - smin) + smin = [(r - 50) / 100] \times 255$$

Computed transformed values for representative pixel intensities across the original range:

r (original)	50	75	100	125	150
s (stretched)	0.0	63.8	127.5	191.3	255.0
s (rounded)	0	64	128	191	255

Interpretation

- The original 100-level range (50–150) is expanded to fill the entire 0–255 range: the darkest original pixel (50) maps to pure black (0) and the brightest (150) maps to pure white (255).
- Every intermediate value is scaled proportionally (a 2.55× expansion factor), so intensity differences that were barely distinguishable in the original image become clearly visible after stretching.
- This significantly improves visual contrast and makes the image far more suitable for visual inspection or for subsequent processing steps such as thresholding and segmentation.